

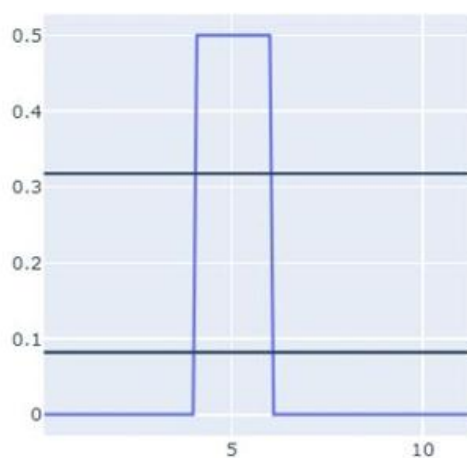
## SLOT-C

## ECA31032- NANO ELECTRONICS

## EXPERIMENT 5 - Demonstration of Quantum Tunneling

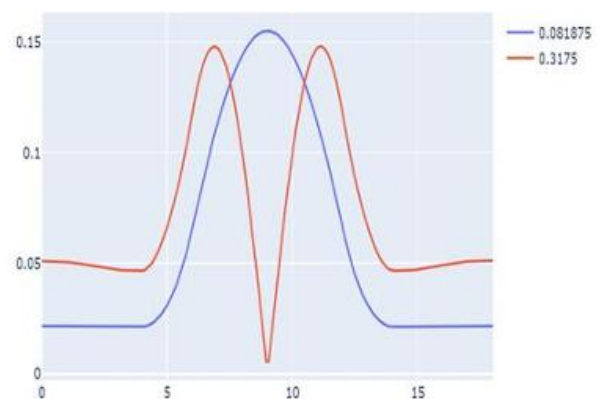
OUTPUT:

Potential Energy vs Distance



f

Wave-functions from tight-binding



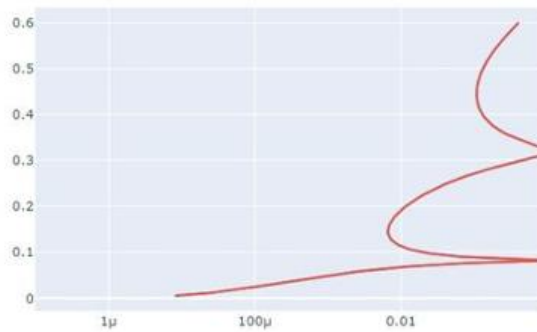
## Explanation

The left graph shows the potential energy profile, where the rectangular barrier represents the region that confines electrons. The horizontal lines indicate different electron energy levels.

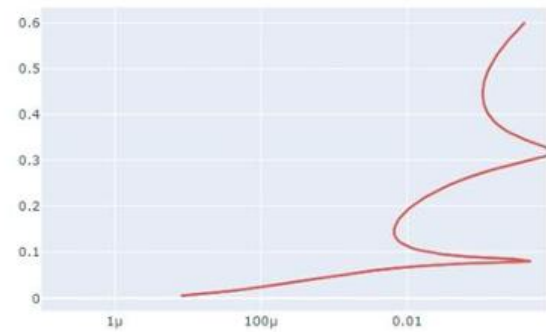
The right graph shows the wave functions calculated using the tight-binding method. The blue and red curves represent different electron energy states, with the peaks indicating a higher probability of finding the electron. Together, these graphs demonstrate quantum confinement and electron behavior in a semiconductor nanostructure.

## CASE STUDY 1: Analysis of Transmission Coefficient vs. Energy on Refined Grid

Transmission Coefficient Vs Energy on refined grid



Transmission Coefficient Vs Energy on Homogeneous Grid

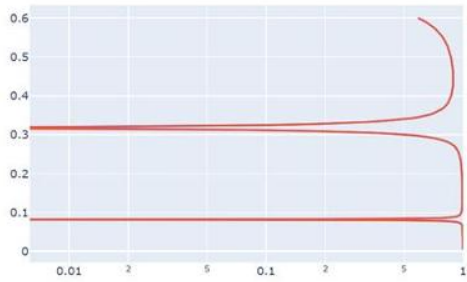


### Explanation

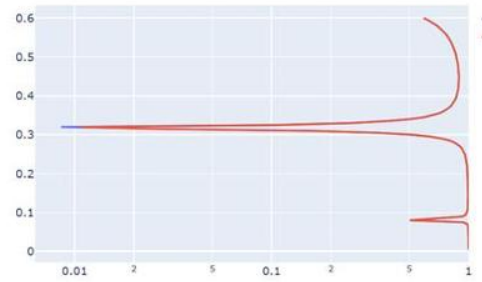
The transmission coefficient represents the probability of a particle passing through a potential barrier. In this simulation, a refined grid is used to obtain more accurate results by dividing the simulation region into smaller intervals. As the particle energy increases, the transmission coefficient generally increases, indicating that particles can pass through the barrier more easily. This analysis helps in understanding quantum tunneling and electron transport in semiconductor nanostructures.

### CASE STUDY 2: Analysis of Reflection Coefficient vs. Energy on Homogeneous Grid

Reflection Coefficient Vs Energy on refined grid



Reflection Coefficient Vs Energy on Homogeneous Grid



## Explanation

The reflection coefficient represents the probability of a particle being reflected by a potential barrier. In this simulation, a homogeneous grid with uniform spacing is used to analyze the reflection behavior. At lower energy levels, the reflection coefficient is higher because particles have insufficient energy to cross the barrier. As the energy increases, the reflection coefficient decreases, indicating improved transmission. This study is useful for analyzing wave behavior and electron scattering in semiconductor and quantum devices.

## Conclusion

The analysis of the Transmission Coefficient vs. Energy on a Refined Grid and the Reflection Coefficient vs. Energy on a Homogeneous Grid demonstrates the quantum transport behavior of particles through a potential barrier. The refined grid provides more accurate transmission characteristics by capturing rapid variations in energy, while the homogeneous grid effectively illustrates the reflection behavior over a uniform spatial distribution. Together, these analyses confirm that the transmission coefficient increases with favorable energy levels, whereas the reflection coefficient decreases, satisfying the conservation of probability ( $\text{Transmission} + \text{Reflection} \approx 1$ ). These simulations are useful for understanding electron transport in quantum wells, nanowires, semiconductor devices, and nanoscale electronic systems.